

Pilot Report — Live Two-Track Forecaster (“Avraa’s Prophet”)

Date: 2026-06-09 (two days before the June 11 kickoff) **Status:** PASSED — the live loop runs end to end and reacts correctly to every feed. **Script:** `scripts/pilot.py` (reusable; edit team / news / score at the top to rehearse any scenario)

1. Objective

Confirm, before any real game is played, that the live forecasting loop works end to end: that a **pre-kickoff news event** moves the *before* forecast, that a **result** moves the frozen track, and that **match performance** moves the learning track — each in the right direction — and that the information content (KL) is measured at each step.

2. Test design

A single fabricated scenario built to exercise all three feeds at once, on real ratings (the 2022 structure as the working rig; the machinery is format-agnostic):

Element	Fabricated value
Team under test	Brazil (pre-tournament favourite)
Fake news (before kickoff)	star striker injured → −80 Elo
Fake result	Brazil 0–2 Serbia (an upset loss)
Fake performance	Brazil out-xG’s Serbia 3.0 to 0.3 (i.e. the loss was a freak)
Monte-Carlo	N = 5,000, seed = 2026

The scenario is deliberately adversarial: the *result* says Brazil collapsed, the *performance* says Brazil dominated. A correct system must treat them differently.

3. Procedure and results

Stage	Brazil champion	What it exercises
A. t = 0 baseline	28.5 %	—
B. + fake injury news (−80 Elo) → <i>before</i> snapshot	15.3 %	the news / before-feed path (−13.2 pts; 0.075 bits of information)
C. + fake game (0–2 loss, out-xG’d 3.0–0.3): → Frozen (conditions on the result only)	8.9 %	the after-feed path, both tracks result conditioning (KL 0.049)
→ Learning (also updates strength from xG)	18.5 %	performance learning (rating 2089 → 2161)

4. Findings

1. **The news / before path works.** The injury moved Brazil’s champion probability from 28.5 % to 15.3 % and registered 0.075 bits of information — exogenous news correctly updates the *before* forecast.
2. **The result path works.** Conditioning on the 0–2 defeat dropped the frozen track to 8.9 %, as it should for a favourite that loses its opener.
3. **The performance path works.** The learning track read the 3.0-vs-0.3 xG, lifted Brazil’s strength (2089 $\rightarrow$ 2161), and held it at 18.5 % — **+9.6 points above the frozen track** — correctly judging the defeat to be unlucky. This is the same behaviour the system showed on Argentina’s real 2022 loss to Saudi Arabia.
4. **The information measure works.** KL divergence was computed and non-zero at each transition (0.075 bits for the news, 0.049 for the game).
5. **Responsiveness observation (the useful surprise).** The learning track ended *higher* than the post-news snapshot (18.5 % > 15.3 %): one dominant performance more than offset both the injury news and the loss, and the drift reached +72 of a ± 75 cap. The *direction* is right, but the system is very sensitive to a single extreme game. This is exactly what the regularisation knob k governs.

5. Verdict and recommendation

The live two-track loop is **flight-tested and ready** for June 11: news, result, and performance each move the forecast correctly, and the frozen-vs-learning contrast behaves as designed.

Recommended next step before kickoff: sweep the learning-rate knob k (and the per-game drift cap) on this scenario to pick a default that is responsive without over-reacting to one freak result — then lock it for the tournament.

6. Reproducibility

`python3 scripts/pilot.py`

Deterministic (seed = 2026). The scenario parameters (team, news delta, result, observed xG) are constants at the top of the script and can be changed to rehearse any event — a useful dry run on the morning of the opening match.